

Dustin J. Basler

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[Google Scholar](#) | [LinkedIn](#)

EDUCATION

B.S. Biomedical Sciences

Southern Illinois University - Carbondale, May 2020

Associates of Science

John A. Logan Community College, May 2016

Herrin High School, Herrin, IL

High School Diploma, June 2015

PROFESSIONAL EXPERIENCE

Electro-Mechanical Technician

Advanced Testing Laboratories, Bureau Veritas – Ethicon, J&J, R&D Digital Surgery (Mar 2024– Present)

- Facilitated medical device testing utilizing robotic tool drivers and ISMs for multiple teams
- Performed testing on human/animal tissue and resected staple lines for processing
- Assisted in Test Methods Development and Validation processes
- Executed Laser Welding procedures and assembly line work for Medical Device production
- Reviewed and Executed GMP, GDP, DI relevant to DV/EB Testing for FDA validation/approval

Project Coordinator II, Medpace, Inc. - Medpace Core Laboratory (Aug 2022– Oct 2023)

- Managed the medical imagery (MRI, ECG, DXA, CT) of 7 clinical trials
- Coordinated trial specific duties for 1000+ sites globally
- Maintained supplies inventory for study sites and organize shipping efforts of study materials
- Scheduled project meetings (internal and external), been an active meeting participant, and provide post meeting documentation/minutes
- Provided insightful input on project-related documents (essential document collection process, final analysis plan, final study report, etc.)

Research Assistant II

(Jun 2020 – Aug 2022)

Center for Pulmonary Imaging Research, Cincinnati Children's Hospital

- Prepared hyperpolarized xenon-129 gas and conducted data analysis for 150+ patients undergoing xenon MRI
- Scheduled and managed documentation records for 30 research studies
- In-charge of billing and accounting on 20+ research grants
- Lab manager and safety officer for 3 laboratory spaces (inventory, personnel, upkeep)
- Improved and developed standard operating procedures for laboratory equipment

Undergraduate Research Assistant, Southern Illinois University Carbondale (Aug 2016 – Jun 2020)

- Prepared hyperpolarized xenon-129 and xenon-131 gas using spin-exchange optical pumping (SEOP) with high-powered lasers
- Performed optically detected electron spin resonance (ODESR) to characterize the electron spin polarization of alkali metal vapors used in SEOP

Scholar, Southern Illinois Bridges to the Baccalaureate

(Jan 2016 – Jun 2018)

- Designed and implemented a research project studying microtubule-associated proteins and their effects on plant morphology over the course of 3 months
- Used microscopy and genetic analysis to characterize extracted and purified DNA from the *Arabidopsis thaliana*
- Created a research-based poster and talk presented to the public

Board Operator/Assistant DJ, Withers Broadcasting, Marion, IL

(Aug 2012 – Aug 2018)

- Coordinated over 100 On-Air local and national advertisements for Southern Illinois Miners baseball team
- Managed broadcasting of 20 local businesses and high school sports over 4 seasons

SKILLS

- All aspects of preparing and performing SEOP experiments, including preparation of SEOP cells (handling alkali metals, coatings, glassware, cleaning), high-vacuum / pressure gas handling / manifolds, high power laser operation (optics, power measurements, near-IR spectroscopy, PSUs, temperature control, output characterization), low-field *in situ* NMR / polarimetry, and simple RF circuitry
- Data processing / software (including Origin, MestreNova, Magritek Prospa [Aurora, Kea], SpectraSuite, MATLAB, ITK Snap, Image J, Microsoft Suite, SolidWorks)
- Experienced with dangerous goods (DG) international shipments and protocols
- Refined written and oral communication skills through research-specific presentations and report writing.
- Effectively conveys complex ideas with simple examples in the following areas: chemistry, physics, physiology, philosophy, and technology
- Strong critical thinking and analytical skills; comfortable with abstract conceptual arguments, quantitative analysis, graphical modeling, statistics, and field-specific language
- Well-developed organizational skills; including laboratory notebook keeping, scheduling, and file logging
- Accountancy & Lab Management: Developed and organized budgets for Clinical Research, R&D

CERTIFICATIONS

- Six Sigma Black Belt Certification (In Progress – Self Paced)

RECOGNITIONS

- “Living the Values” Award – BV ATL (Nov 2024)
- SIU Alumni Association Undergraduate Research Assistantship (Aug 2018 – May 2019)
- Southern Illinois Bridges to the Baccalaureate Scholar (Jan 2016 – Aug 2016)
- Future Business Leaders of America Conference Runner-Up in Accounting (Feb 2015)

CLINICAL SHADOWING

Cincinnati Children’s Hospital Medical Center

- Dr. Ajay Kaul, MD – Director, Neurogastroenterology | ajay.kaul@cchmc.org
- Dr. Gregory Burg, MD – Internal Medicine | gregory.burg@cchmc.org
- Dr. Erik Hysinger, MD – Pulmonary Physiologist | erik.hysinger@cchmc.org
- Dr. Christopher Towe, MD – Rare Lung Disease | christopher.towe@cchmc.org

VOLUNTEERING

- US Air Force ROTC, Det. 205, Southern Illinois University Carbondale (Aug 2015 – Feb 2019)
- Council Representative - Republican Party- City of Marion, IL (Aug 2018 – Aug 2019)
- Trail Building Volunteer – Touch of Nature, Carbondale, IL (May 2016 – Aug 2016)
- Volunteer - Keep Carbondale Beautiful (May 2016 – Aug 2016)

PEER-REVIEWED PUBLICATIONS

[2024](#) H Lu, MJ Barlow, **D Basler**, P Gutfreund, O Holderer, A Ioffe, S Pasini, P Pistel, Z Salhi, K Zhernenkov, BM Goodson, WM Snow, E Babcock. “First measurement of neutron birefringence in polarized ^{129}Xe and ^{131}Xe nuclei” in *Physical Review C*

[2023](#) Michael J Molway, Liana Bales-Shaffer, Kaili Ranta, James Ball, Eleanor Sparling, Mia Prince, Daniel Cocking, **Dustin Basler**, Megan Murphy, Bryce E Kidd, Abdulbasit Tobi Gafar, Justin Porter, Kierstyn Albin, Matthew S Rosen, Eduard Y Chekmenov, W Michael Snow, Michael J Barlow, Boyd M Goodson “Dramatic improvement in

the “Bulk” hyperpolarization of ^{131}Xe via spin exchange optical pumping probed using in situ low-field NMR”. Submitted to *Journal of Magnetic Resonance*.

2021 Michael J Molway, Liana Bales-Shaffer, Kaili Ranta, **Dustin Basler**, Megan Murphy, Bryce E Kidd, Abdulbasit Tobi Gafar, Justin Porter, Kierstyn Albin, Boyd M Goodson, Eduard Y Chekmenev, Matthew S Rosen, W Michael Snow, James Ball, Eleanor Sparling, Mia Prince, Daniel Cocking, Michael J Barlow “Two-Orders-of-Magnitude Improvement in the Total Spin Angular Momentum of ^{131}Xe Nuclei Using Spin Exchange Optical Pumping”. submitted to *arXiv:2105.03076*.

CONFERENCE ABSTRACTS

2022 **Dustin J. Basler**, Abdullah S. Bdaiwi, Laura L. Walkup, Matt Willmering, Zackary I. Cleveland, Jason C. Woods. “Texture Analysis as an Adjunct to Hyperpolarized ^{129}Xe Ventilation Defect Percentage for Quantifying Cystic Fibrosis Lung Disease Severity” 31st International Society for Magnetic Resonance in Imaging (ISMRM), London, UK, May 2022 (poster).

2020 J. Porter, **D. Basler**; J. Plummer, G. Russell, B. Kidd, E. Chekmenev, M. J. Barlow, B. M. Goodson. “Optically Detected ^{85}Rb / ^{133}Cs Electron Spin Resonance (ODESR) in “Hybrid” Spin-Exchange Optical Pumping Cells with Xenon Density Variance” 61st Experimental Nucl. Magn. Reson. Conference, Baltimore, MD, March 2020 (poster).

2020 **D. Basler**, K. Albin, L. Bales, A.T. Gafar, H Lu, W. M. Snow, E Babcock, E. Y. Chekmenev, M.J. Barlow, and B.M. Goodson. “NMR of ^{131}Xe Spin Exchange Optical Pumping in Aluminosilicate (GE180) Cells: Targets for Neutron Optics Searches for Time-Reversal Invariance Violation.” 61st Experimental Nucl. Magn. Reson. Conference, Baltimore, MD, March 2020 (poster).

2019 Porter J, **Basler D**, Plummer JW, Russell G, Kidd BE, Chekmenev EY, Barlow MJ, Goodson BM. “ ^{85}Rb / ^{133}Cs Optically Detected Electron Spin Resonance in “Hybrid” Spin-Exchange Optical Pumping Cells in Clinically Relevant Regimes.” 60th Experimental Nucl. Magn. Reson. Conference, Asilomar, CA, April 2019 (poster).

2019 L. Bales, **D. Basler**, K. Albin, S. Islam, T. Gafar, J. Ball, E. Sparling, M.J. Barlow, E.Y. Chekmenev, and B.M. Goodson, “Developing Hyperpolarized ^{131}Xe as a Potential Target for Neutron Optics Searches for Time-Reversal Invariance Violations.” 60th Experimental Nucl. Magn. Reson. Conference, Asilomar, CA, April 2019 (poster).

2018 BM Goodson, K Albin, L Bales, MJ Barlow, **D Basler**, EY Chekmenev, A Coffey, T Gebhardt, ME Gemeinhardt, JL Gesiorski, Y Hou, RK Irwin, BE Kidd, IV Koptuyug, KV Kovtunov, T LaGambina, MN Limbach, JA Mashni, MJ Molway, M Murphy, P Nikolaou, J Porter, MS Rosen, RV Shchepin, F Shi. “From Cleavable ‘Double Agents’ to Polarized Targets: New Systems and Approaches for SABRE and SEOP Hyperpolarization.” *HYP18*, Southampton, UK, September 2-5, 2018 (poster).

2018 Michael J. Molway, Bryce E. Kidd, Liana Bales, Megan Murphy, **Dustin Basler**, Kierstyn Albin, Michael J. Barlow, Eduard Y. Chekmenev, and Boyd M. Goodson “Probing $\text{Cs}/^{131}\text{Xe}$ versus $\text{Rb}/^{131}\text{Xe}$ Spin Exchange Using in situ Hyperpolarized ^{131}Xe NMR.” 59th *Experimental Nucl. Magn. Reson. Conference*, Orlando, FL, April 30-May 4, 2018 (poster).

2018 Michael J. Molway, Bryce E. Kidd, Liana Bales, Megan N. Murphy, **Dustin Basler**, Justin Porter, Panayiotis Nikolaou, Aaron Coffey, Matthew S. Rosen, Michael J. Barlow, Eduard Y. Chekmenev, and Boyd M. Goodson “Enhancing SEOP Hyperpolarization of ^{129}Xe : Alkali Metal Choice, Laser Technology, and Cell Design.” 59th *Experimental Nucl. Magn. Reson. Conference*, Orlando, FL, April 30-May 4, 2018 (poster).

2018 Bryce E. Kidd, Justin Porter, **Dustin Basler**, Robert K. Irwin, Tania LaGambina, Eduard Y. Chekmenev, Michael J. Barlow, and Boyd M. Goodson “Optically Detected ESR of ^{85}Rb & ^{133}Cs during Clinical-Scale Stopped-Flow Spin-Exchange Optical Pumping of Xe.” 59th *Experimental Nucl. Magn. Reson. Conference*, Orlando, FL, April 30-May 4, 2018 (poster).

2017 Megan N. Murphy, Justin Porter, Bryce E. Kidd, Michael J. Molway, **Dustin Basler**, Kaili Ranta, Max E. Gemeinhardt, Aaron M. Coffey, Panayiotis Nikolaou, Bamidele D. Falola, Ian I. Suni, Matthew S. Rosen, Michael J. Barlow, Eduard Y. Chekmenev, and Boyd M. Goodson. “New Developments in Stopped-Flow Spin-Exchange Optical Pumping of ^{129}Xe and ^{131}Xe ”, 58th *Experimental Nucl. Magn. Reson. Conference*, Pacific Grove, CA, March 26-31, 2017 (poster).